

ELECTIVECOURSE-4.3

UME715: BATTERY MANAGEMENT SYSTEMS				
	L	T	P	Cr
	2	2	0	3.0
Course Objectives: The objective of this course is to provide a comprehensive understanding of Battery Management Systems (BMS), modeling and simulating battery packs, state and health estimation of batteries using Kalman filters, cell-balancing techniques, and voltage-based power-limit estimation, focusing on practical applications in electric vehicle battery pack design and optimization.				
Introduction to Battery-Management-System Requirements: Introduction and BMS functionality, requirement of sensing, requirement of high-voltage contactor control, requirement of isolation sensing and thermal control, requirements of protection and interface, requirements of state-of-charge estimation, requirements of energy and power estimation. Modeling Battery Packs: Modeling approach: equivalent-circuit models, physics-based models; simulating an electric vehicle; equations for vehicle dynamics; vehicle range calculations, examples; simulating constant power and voltage; simulating battery packs. Battery State Estimation: Preliminary definitions, approaches to estimate the state of charge, deriving the linear Kalman filter, visualizing the Kalman filter, practical considerations; the extended Kalman filter (EKF), implementing EKF on ESC model, problems with EKF, improvements with sigma-point methods, the SPKF steps, implementing SPKF on ESC model; real-world issues pertaining to sensors, initialization: real-world issues: speed, solved by “bar-delta” filtering, bar-delta filtering using the ESC cell model. Battery Health Estimation: Lithium-ion aging; negative electrode and positive electrode; sensitivity of voltage to ESR and total capacity; a Kalman filter framework for estimating parameters; EKF for parameter estimation; simultaneous state and parameter estimation; robustness and speed. Cell Balancing: Causes of imbalance, design choices when implementing balancing; circuits for balancing: passive, active (capacitive, inductive, and DC-DC); requirement of quick balancing of a battery pack. Voltage-Based Power-Limit Estimation: Introduction to voltage-based power-limit estimation, voltage-based rate limits using a simple cell model; voltage-based rate limits using a comprehensive cell model, bisection search, power-limits estimation example.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Electric Vehicle Battery Pack Design: Battery sizing and selection for automotive applications: delve into the methodology behind sizing batteries and choosing the optimal battery for use in automotive contexts, considering various factors such as energy requirements and vehicle specifications. Components and design of battery packs: explore the intricacies of both high and low-voltage components within battery packs, focusing on battery cell design.

Course Learning Outcomes (CLOs)

After completion of this course, students will be able to:

1. understand the core requirements and functionalities of BMS in mobility systems.
2. model and simulate battery packs using various modeling approaches and vehicle dynamics equations.
3. evaluate battery state and health estimation techniques, including Kalman filters and sigma-point methods.
4. design cell balancing techniques and voltage-based power-limit estimation for automotive battery packs.

Text Books

1. Plett, Gregory L. Battery management systems, Volume II: Equivalent-circuit methods. Artech House, 2015.
2. Bergveld, H.J., Kruijt, W.S., Notten, P.H.L “Battery Management Systems -Design by Modelling” Philips Research Book Series 2002.
3. Davide Andrea,” Battery Management Systems for Large Lithium-ion Battery Packs” Artech House, 2010.

Reference Books

1. Plett, Gregory L. Battery management systems, Volume I: Battery modeling. Artech House, 2015.
2. Pop, Valer, et. al. Battery management systems: Accurate state-of-charge indication for battery-powered applications. Vol. 9. Springer Science & Business Media, 2008.
3. Weicker, P. A systems approach to lithium-ion battery management. Artech House, 2013.

Evaluation Scheme

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
	Evaluation Elements	Weightage (%)

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